

1.3 Tasks

- Data Loading: Before starting this task please refer to weeks seven to nine seminars
 - Extract and store the S&P 500 Tickers from 'https://en.wikipedia.org/wiki/List_of_S%26P_500_companies' using Pandas
 - Using 'yfinance' & 'YahooFinancials' packages load the S&P 500 Tickers daily market performance as a Pandas data frame for the period 01-01-2022 to 01-01-2025
 - Identify and remove invalid data

Define and calculate in Google Colab the Daily Return, Beta and Annual Volatility Metrics: Before starting this task please refer to week 7-9 seminars.

- **Daily Return=**
 - **$Beta = (correlation\ of\ stock's\ returns\ and\ index's\ returns) * (Stock's\ standard\ deviation\ of\ returns / index's\ standard\ deviation\ of\ returns)$**
 - **Annual Volatility = Daily Return Standard Deviation**
 - Illustrate the use of the Daily Return, Beta and Annual Volatility Metrics in segmenting stocks into categories to help investors align their portfolios with specific risk tolerances and markets.
- Customer segmentation:

Stock segmentation with **agglomerative clustering**: The third step is to divide the stock portfolio into groups for the dimensions (**Daily Return, Beta and Annual Volatility Metrics**), using clustering using **agglomerative clustering**. The report should illustrate the following:

 - **Brief discussion of the appropriateness of agglomerative clustering as the adopted clustering methodology.**
 - Brief discussion the techniques applied to identify the best value of N-number of clusters.
 - Implementation of **agglomerative clustering** using Python via Google Colab.

Stock segmentation with k-means: The fourth step is to divide the stock portfolio into tiered groups for each of the three dimensions (**Daily Return, Beta and Annual Volatility Metrics**), using clustering using K-means and discuss the profile of each found cluster (in terms of the **Beta and Annual Volatility Metrics** that describe the properties of the stocks of each cluster). The report should illustrate the following:

- **Brief discussion of the appropriateness of K-means as the adopted clustering methodology.**
- It is necessary to discuss the techniques applied to identify the best value of K-number of clusters.
- Implementation of K-Means using Python via Google Colab.